Case Study: The Balancing Act

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Overview

The Balancing Act is a self-paced e-learning module that teaches Grade 7–8 students to solve two-step equations, paired with an analytics dashboard that measures whether it worked. It is a sample portfolio project: the module is fully functional, and the results data is synthetic.

Item	Detail
My role	Instructional designer, e-learning developer and data analyst
Audience	Grade 7–8 mathematics students
Duration	About 15 minutes, self-paced
Deliverables	Interactive module and learning analytics dashboard
Tools	HTML, CSS and JavaScript for the module; data analysis and d3 charts for the dashboard

The challenge

In the classroom, linear equations are where many students stop understanding and start memorising. They learn "move it to the other side" as a rule, then apply it wrongly under pressure.

- The most common error is undoing the constant with the wrong operation, for example adding 5 instead of subtracting it.
- Students rarely check their answer by substituting it back.
- Printed worksheets give feedback days later, after the misconception has set in.

Approach and design decisions

The module replaces the rule with a model: an equation is a balanced scale, and whatever you do to one side you do to the other.

- **Concrete before abstract.** Students move x tiles and unit coins on a scale before working with symbols alone.
- **Gradual release.** The lesson runs Learn, Explore, Practice, Check: a worked example, a guided activity, independent problems, then a short assessment.
- **Feedback that teaches.** Every wrong choice gets its own explanation. A practice answer that comes from adding instead of subtracting is recognised and named.
- **Checking as a habit.** Every correct answer shows the substitution that proves it.
- **Built to be measured.** Attempts, hints and quiz scores are the events an analytics dashboard needs.

Development

The module is a single responsive web page that works on phones and laptops, in light and dark mode.

1. **Learn:** learning objectives, the two-step rule and a worked example ending with a check.
2. **Explore:** an interactive balance scale for 3x + 4 = 19. Correct moves update the scale and the equation; a one-sided move visibly tips it.
3. **Practice:** randomly generated problems with whole-number answers, two levels of hints, error-specific feedback and a running tally.
4. **Check:** five multiple-choice questions, including a word problem and an error-analysis item, with explanations and a final score.

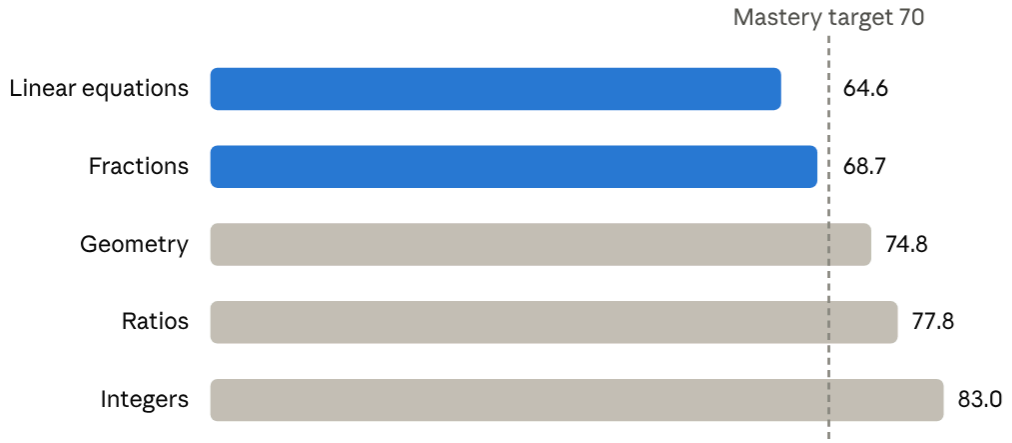
The live module can be opened and tried.

Measuring impact

In the sample data, average scores rose from 56.0 on the pre-test to 75.8 after the module, a gain of 19.8 points. The analytics dashboard breaks this down by class, topic and completion.

Two of five topics fall below the mastery target

Average post-module score out of 100 · 2 topics under 70



Synthetic sample data · 120 students, four classes · computed in the analytics dashboard

- Students who completed 90% or more of the module gained 21.7 points, against 18.7 for those under 70%.
- 16 students still scored below 60 and are listed for small-group support, each with their weakest topic.
- The data points the next design cycle at linear equations and fractions.

Lessons and next steps

- Design and measurement belong together: deciding which events to track while designing the module made the dashboard possible.
- Error-specific feedback is the most valuable part of the module and the hardest to write; it draws directly on classroom experience.
- Next: pilot the module with a real class, replace the synthetic data with real results, and add a fractions module built the same way.